515.31.07 Payment Adjustment - Payment adjustments resulting from the application of this Section will be effected on each progress payment as follows.

1. For each Lot paid for by the square metre (m2), the applicable payment adjustment derived from Table 515-C (Density), in dollars per square metre will be expressed as positive in the case of increases and negative in the case of decreases. The algebraic sum of these unit adjustments will then be applied to the payment quantity for the Lot. The resulting amount, in dollars, will be the net payment adjustment, positive or negative, for that Lot.

The algebraic sum of the net payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for density for the current progress payment.

2. For each Lot, the applicable payment adjustment derived from Table 515-D (Smoothness), in dollars per Lot, will be expressed as positive in the case of increases and negative in the case of decreases, and will be the payment adjustment, positive or negative, for that Lot.

The algebraic sum of the payment adjustments for all such Lots for which payment is authorized on the current progress payment, computed in dollars, shall be the total payment adjustment for smoothness for the current progress payment.

3. The algebraic sum of the total payment adjustments for density and smoothness, derived in accordance with 1 and 2 above, shall be the total payment adjustment, positive or negative, in dollars, for all attributes for the current progress payment. This amount shall be added, if positive, or subtracted, if negative, by a single entry in computing the current progress payment.

4. The process set out in 1 to 3 above will be used in computing each progress payment to which it is applicable.

515.32 Surplus Aggregate In Stockpile – Surplus aggregate in stockpile in Ministry pits shall be the property of the Ministry.

515.32.01 Ministry Purchase of Surplus Aggregate – – No additional compensation will be paid for surplus aggregates.

APPENDIX 1

QUALITY CONTROL REQUIREMENTS AND GUIDELINES

1.01 GENERAL

The Contractor's responsibilities for preparing, submitting and adhering to a Quality Control Plan are specified in DB SS 515.04. This Appendix provides requirements and guidelines for the Contractor's Quality Control Plan, in addition to those set out in DB SS 515.04.

1.02 QUALITY CONTROL PLAN

As a requirement, the Contractor shall prepare and submit a Quality Control Plan for evaluation. The Plan may be operated wholly, or in part by a qualified Subcontractor or an independent organization/agency. However, the Quality Control Plan, including compliance with the Plan and its modifications, is the responsibility of the Contractor.

The Plan shall address all elements that affect the quality of the hot in place recycled asphalt pavement, including but not limited to the following:

- Supply of Asphalt Materials
- Supply of rejuvenating agents
- Identification of source and proof of quality of virgin aggregates to be supplied;
- Virgin aggregate production and its gradation control;
- Quality of virgin aggregate components;
- Stockpile management;
- Asphalt plant calibration and proportioning of aggregate components;
- Mixing including asphalt cement content control;
- Process temperature controls;
- Admix Material and Rejuvenating Agent application rates and processed dimensions, placing and finishing;
- Joints;
- Density;
- Smoothness;
- Segregation;

The plan shall also include the following:

- The name of the Quality Control testing agency and its proven capability to provide the specific services required for the project.
- The list of dedicated technical staff, if available, (including names, qualifications and relevant experience) and their proposed roles.

The list of testing equipment available for project work.

The Quality Control Plan shall include the designation of specific personnel responsible for specific quality control duties.

- There shall be a designated Quality Control Manager, as per DB SS 515.04.03. The Quality Control Manager shall be qualified as per DB SS 515.04.03. The Quality Control Manager shall be responsible for the preparation and sign off of the Quality Control Plan, responsible for all Quality Control testing, and inspections, responsible for the sign off of all quality control testing and inspection records and responsible for all quality control submissions to the Ministry. The individual shall be identified and named in the Quality Control Plan submission. If the Contractor requests a change in the Quality Control Manager, then a formal request to the Ministry Representative shall be made and the Ministry will review the request for approval.
- There may be a designated Process Control Technician (PCT) who would be responsible to ensure that laboratory test results and other quality control practices are used to control the quality of aggregates and other mix components and to adjust and control mix proportions to meet the mix design(s). The PCT is responsible for ensuring that testing equipment, utilized for proportioning and mixing are calibrated and in good working order. The Plan may describe how the PCT's duties, including sampling methods and responsibilities are to be accomplished and documented. The Plan should also describe the criteria to be used by the PCT to correct or reject unsatisfactory materials.
- There may also be a Pavement Quality Control Technician (PQT) who would be responsible to ensure that delivered materials meet the requirements of the specifications. In addition, the PQT shall be responsible for periodically inspecting all equipment used in recycling, placing, finishing, and compacting to assure its proper operating condition and to assure that recycling, placing, application rate, finishing, joint construction, and compaction is in conformance with this specification and the contract requirements.

1.03 CONTRACTOR'S RECORD OF QUALITY CONTROL TESTING

Test results shall be made on specified forms or charts immediately after completion of each test. These test results are to be made available to the Ministry Representative upon request.

Records of gradation control, both during aggregate production and also during the asphalt mixing operation, should be kept on the form H-295 - Mechanical Analysis of Aggregates.

1.04 MATERIALS APPLICATION RATE

The Contractor shall control the Material Application Rate by monitoring the amount of asphalt admix delivered to the road against the area covered by checking the application rate every five loads.

The Contractor shall monitor the addition of rejuvenating agents by calculating the application rate, through comparing the metered amount with the total area processed every one-half hour.

The Contractor is to advise the Ministry Representative on an ongoing basis of the application rates.

1.05 DENSITY

The Contractor shall take core samples to determine actual pavement density. At the start of paving, the Contractor may take a minimum of two pavement cores from each Sub-Lot. The Contractor may employ a nuclear densometer to ensure intermediate density control.

1.06 OTHER QUALITY CONTROL PROCEDURES

The Contractor may initiate other Quality Control procedures as necessary for ensuring production of a quality product and include them in the Quality Control Plan. Procedures may also be introduced after the start of work as necessary as amendments to the Quality Control Plan.

1.07 QUALITY CONTROL TESTING FREQUENCY GUIDELINES

Test frequency guidelines for Quality Control are described in TABLE 1:

TABLE 1 –TEST FREQUENCY GUIDELINES

	Test	Minimum Frequency
Tests During Crushing (Virgin Aggregate)	ASTM C-136, Dry Sieve Analysis of Aggregate	Split Stockpiles: One for each stockpile for every 2 hours of production. One main stockpile: for every 300 tonnes. Blend Sand: One for every 100 tonnes during stockpiling. Natural filler: One for every 50 tonnes during stockpiling.
	ASTM D-5821 Determining the Percentage of Fractured Particles in Coarse Aggregate	Every second coarse aggregate sieve test.
	ASTM C-117 Sieve Analysis of Aggregates by Washing (Field Lab)	One per day on reduced sample obtained from combined samples from the crusher
Tests During Asphalt Plant Mixing (Admix)	ASTM C-136, Dry Sieve Analysis of Aggregate	One of combined aggregate every 100 tonnes.
	ASTM C-566 & D-2216, Moisture Content	Aggregate: 2 tests/Lot Asphalt mix: 1 on first Sub-Lot and every second day.
	ASTM C-117 Sieve Analysis of Aggregates by Washing (Field Lab)	One per shift on reduced sample obtained from combined samples from the plant cold feed
	Asphalt Extraction Test ASTM D-6307 Ignition Method	One per Lot.
	Penetration of Bituminous Materials ASTM D-5	One per Manufacturer's Batch
	Viscosity ASTM D-2171	Contractor's Option
Rejuvenating Agents Tests	Rejuvenating Agent	Contractor's option.
Test During Asphalt Paving for Density Testing (Hot In Place Recycled Pavement)	ASTM D-1559, Resistance to Plastic Flow of Bituminous Mixtures Using Marshall Apparatus	Contractor's Option
	ASTM D-2726 Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures (Briquette or Cores)	One per Sub-lot
	ASTM D-2041 Theoretical Maximum Specific Gravity of Bituminous Paving Mixtures	One (see Appendix 3 for Combining Cores)
	Core Samples	One core for each Sub-Lot. All cores to be 150mm diameter.

APPENDIX 2

REQUIREMENT FOR PRODUCTION OF ADMIX AGGREGATES

2.01 GRADATION MAXIMUM PERMISSIBLE VARIATION LIMITS:

Sieve Size	
19.0 mm or 16.0 mm	0%
12.5 mm	+/-3.5%
9.5 mm	+/-3.5%
4.75 mm	+/-3.0%
2.36 mm	+/-3.0%
1.18 mm	+/-2.0%
0.600 mm	+/-2.0%
0.300 mm	+/-1.5%
0.150 mm	+/-1.0%
0.075 mm	+/-0.75%

The variation limits, when applied to the Ministry specified design Admix gradation, shall not result in a gradation that is outside of the limits as specified by the Ministry.

APPENDIX 3

METHOD TO COMBINE ROAD CORES INTO A SINGLE COMBINED SAMPLE FOR THE DETERMINATION OF MAXIMUM THEORETICAL DENSITY.

3.01 INTRODUCTION

After the individual Density for each core has been determined, the Sub-lot road cores shall be combined into one single sample for the determination of maximum theoretical density.

3.02 SAMPLE PREPARATION

The upper re-cycled portion of the cores shall be separated from other pavement layers by sawing or other effective methods.

The minimum combined sample size required shall be 2000 grams and 2500 grams for 19.0 mm and 25.0 mm maximum aggregate size respectively.

Select a core trimmer to ensure that the minimum sample size is obtained depending on the thickness of the samples. 100 mm, 114 mm, or 127 mm core trimmers may be used.

Place the core samples in a pan and heat to 130° C ± 5°C for about 20 minutes to allow the specimens to be easily trimmed. Do not overheat. The cores can be stacked on top of each other and trimmed together or individually.

Select a core trimmer to attain the minimum combined sample size. Heat the base of the trimmer by placing on a burner for a few minutes.

Centre the heated trimmer on the heated cores (allow equal clearance on all sides of the core).

Hold the trimmer vertically by its handle, then press it down through the core until it reaches the bottom of the pan. Apply a slight twist if the heated cores(s) are difficult to penetrate.

Remove the outside cut rock mix by means of a spatula spoon and discard.

Thoroughly mix the resultant combined sample prior to testing to ensure uniform aggregate coating.

The complete combined sample must be tested. If the resultant sample size exceeds the available capacity of the container used for vacuum saturation, it may be tested a portion at a time.

SECTION 521

ASPHALT STABILIZED BASE COURSE MIXED IN PLACE

DESCRIPTION

521.01 Scope - This Section describes the materials, plant, equipment and work practice required for the in place stabilization of granular base course material mixed with selected bituminous binder.

This product may only be used if explicitly authorized in the Contract or in the sole discretion of the Ministry Manager

EQUIPMENT

521.21 General Condition of Plant and Equipment

DBSS 521.22 to DBSS 521.25 inclusive are guidelines only, and as such, are not mandatory.

521.21.01 Not Used

521.21.02 Not Used

521.21.03 All equipment and plant shall be in good mechanical condition and adjustment and be capable of efficiently performing the work as required by the Contract.

521.21.04 Not Used

521.21.05 The Ministry Representative shall have access at all times to any plant, equipment or machinery to be used on this contract in order to check calibration, control or operating adjustments

521.21.06 Not Used

521.21.07 Not Used

521.22 Pressure Distributor - The pressure distributor used for applying bituminous materials may consist of a fully insulated tank with a minimum capacity of 5000 ℓ permanently and rigidly mounted on a truck or trailer provided with pneumatic tires and propelled by a power unit capable of maintaining a constant and uniform speed. The distributor may be provided with the following minimum equipment:

a) quick opening gate in the dome.

b) a measuring dipstick calibrated in 100 ℓ increments or in 20 mm gradations and provided with a calibration chart indicating the number of litres for each 20 mm in depth of contained bituminous material for accurate checking of the distributor contents. In addition an external and readily visible depth gauge or levelometer indicating the asphalt content of the tank to the nearest 100 ℓ and having an accuracy of ± 50ℓ**.**

c) an efficient and positive means of heating the bituminous contents uniformly to any temperature up to 175°C and maintaining the contents constantly at any selected temperature without any local overheating.

d) an approved industrial type thermometer so situated that its face shall always be readily visible to the spray bar operator. The sensing element of the thermometer shall be placed either directly in contact with the hot bituminous material or enclosed in a thin walled well of approved manufacture, so placed as to accurately measure the temperature of the material in the tank. The thermometer shall have a minimum range of 10° to 175°C subdivided into 1°C graduations and be accurate to ± 0.5°C

e) rear mounted spray bars and nozzles set parallel to the surface to be sprayed and capable of vertical and lateral adjustment. The spray bars shall be capable of adjustment to provide a minimum spraying width of 1.8 m and a maximum spraying width of 7.5 m.

The feed to the spray bars may be so arranged that there is be a constant circulation of hot bituminous material to the bars such as will ensure a uniform viscosity and constant pressure of the bituminous material at each nozzle, both before and during spraying operations. Feed manifolds to spray bars may be provided with strainers to prevent clogging of bars and nozzles.

f) rotary adjustable, spray bar nozzles which may be so designed and set as to ensure a uniformly fan shaped spray without atomization, overlapping on the surface to be sprayed, such that there is a uniformly sprayed surface to clearly defined edges of the sprayed lane without any reduction of the rate of application adjacent to the edges of the lane. Spray nozzles may be provided with valves capable of instant full opening or positive cut-off.

g) a connection to a circulating pressure manifold to which a hose may be attached for a single nozzle, hand operated outlet.

h) a pressure gauge accurate to 15 kPa and calibrated pressure pump, equipped with a tachometer registering litres per minute and driven by an independent mechanically governed power unit, so designed as to be capable of applying through the spray nozzles fitted, accurately measured quantities of bituminous material at a minimum rate of 900 ℓ per minute.

Both pressure gauge and tachometer should be mounted so as to be readily visible to the operator.

i) a fifth wheel tachometer, registering metres per minute and so calibrated and mounted as to be readily visible to the driver of the distributor, to enable the operator to maintain the constant speed required for application of bituminous material to the road at the specified rate.

521.23 Blade Graders - Self-propelled blade graders mounted on pneumatic tires may be of approved modern type, either four wheel drive or tandem type having sufficient capacity to efficiently perform all work that may be required on the contract

521.24 Pulverizing and Rotary Mixing Machines -

TABLE 521-A CRITERIA FOR BLADE GRADERS

Minimum Weight	7.25 t
Minimum Blade Length	3.6 m
Minimum Wheel Base	5.5 m
Minimum Cubic Displacement of Pistons: 4 Cycle Motors	7.0 ℓ
Minimum Cubic Displacement of Pistons: 2 Cycle Motors	4.0 ℓ

Pulverizing and rotary mixing machines to be used on the work may be standard machines, factory produced by a reputable manufacturer, so designed and constructed as to satisfactorily break up all base aggregations into constituent particles and capable of thoroughly and uniformly mixing the granular base material and applied asphaltic binder so as to produce a uniformly coated mix free from lumps, balls or segregation

521.25 Compaction Equipment - Compaction equipment may consist of steel wheel rollers, pneumatic tired rollers, vibrating rollers or tampers.

521.25.01 Self Propelled Steel Rollers - These rollers may be equipped with power units of not less than four cylinders , and under working conditions be capable of developing a compression in the rear wheels of not less than 4.0 kg/mm² over a minimum roller width of 500 mm, with one roller capable of developing a roll pressure of 4.5 kg/mm² of width. Rollers should be in good working condition and free from backlash, faulty steering mechanism or worn parts. Rollers may be equipped with adjustable scrapers to keep the rolls clean and with an efficient means of keeping the wheels wet to prevent mixes from sticking to the rolls. Rolls should be free of flat areas, dents, openings or projections which will mar the surface of the pavement.

521.25.02 Pneumatic Tire Rollers - Pneumatic tire rollers may be either self-propelled or towed type, single or double axle, having a minimum effective rolling width of 1.2 m. The rollers may be equipped with smooth tread pneumatic tires of equal size and diameter. The wheels of the roller should be so spaced that two passes of a single axle roller or one pass of a two axle roller will accomplish one complete coverage equal to the rolling width of the machine. There should be a minimum 6 mm overlap of the tracking wheels of a double axle roller. The wheels of towed rollers to be used on granular base preparation may wobble. The roller should be so constructed that the contact pressure is uniform for all wheels and the tire pressure of the several tires should not vary more than 35 kPa. Pneumatic tire rollers should be constructed with ample ballast space to provide an operating weight of 4.5 to 8 kg/mm² of tire track width. The towing vehicle for towed type roller should be equipped with smooth tread pneumatic tires.

521.25.03 Vibrating Tampers - Vibrating tampers to be used for the compaction of bituminous base in places inaccessible to rollers, should be of a type suitable for the work to be performed.

521.25.04 Vibrating Compaction Equipment - The Contractor may use vibrating compaction equipment to compact the stabilized base, provided that such equipment produces compaction and surface finish equal to that obtainable by using a steel wheel finish roller having a minimum compression roll pressure of 4.5 kg/mm².

CONSTRUCTION

521.31 Method I - When the roadway has been completed in its entirety by others, the following method shall apply:

a) The granular base shall be regraded to true line, grade and cross-section; the top 65 mm of the base shall be loosened by means of scarifying. The lumps of bonded material are to be broken to constituent gravel by harrowing, blading or rotary tilling. The loose material is to be bladed into a windrow, approximately 10 cm in height on both sides of the road. This windrow will act as a dam in case of spillage or the occurrence of rain after the spraying has taken place.

b) The prepared gravel surface shall be primed on the side not obstructed by the windrow, to the outside width required at a rate of 1 to 1.5 ℓ/m², , employing the bituminous binder as required by the Contract and the Design within a spraying temperature range which corresponds to a kinematic viscosity of 150 to 200 mm²/s in the binder.

Priming shall be applied only when the surface is dry or slightly damp, when the air temperature in the shade is not less than 10°C and rising. When the prime has been completely absorbed by the prepared surface, blade

the windrow of loose surface course onto the primed half of the road and prime the untreated half of the surface in a like manner.

Generally, this work shall be governed by the applicable provisions of DB SS 501.31.

c) Immediately prior to application of the bituminous binder, the windrow of loose gravel shall be bladed across the road to a full width uniform cross section. If wet or damp, it shall be first bladed back and forth until dry unless an emulsion is to be used. Cutback bituminous binder shall not be applied until the moisture content has been reduced to 1.5%. In the case of emulsions, mixing may commence when the moisture content of the gravel has been reduced to 4%.
No asphaltic binder shall be sprayed on the granular aggregate unless the average temperature of the aggregate on the road is 13°C or higher and such temperature can be maintained in the aggregate until the full mixing cycle has been completed.

Upon the layer of graded aggregate, the bituminous binder shall be applied uniformly at a rate of 1.35 ℓ/m², at application temperature required to produce a kinematic viscosity of 150 to 200 mm²/s in the binder.

A disc harrow or rotary type mixer shall immediately follow the distributor after the application of the bituminous binder and shall continue to operate on the treated strip until all free bituminous material has been mixed into the mineral aggregate. The treated mixture shall then be further mixed with a blade grader until a thoroughly uniform mixture results.

The mixture shall then be spread uniformly across the road once more, a second uniform application of bituminous binder at a rate of 0.9 to 1.8 ℓ/m², shall be sprayed at application temperature so as to give a final asphalt content of 4 to 6% of the mix. The second application of bituminous binder shall be mixed with the mineral aggregate in a similar manner to the first application, taking care that during the blade mixing none of the untreated base below the prime or shoulder material is brought into the mix. Blade mixing shall be continued until the bituminous binder and mineral aggregates are thoroughly mixed to uniform colour, free from fat spots, balls and uncoated particles. If the mixture should become wet before the mixing process is complete, the mixing operations shall be continued until it has dried out. After final mixing, the mixture shall be brought to a single windrow.

d) Before the finished mixture is finally spread for compaction, a triangular cut shall be made with a blade at each edge of the base course to provide for a thickened edge of bituminous mixture. The cut shall be approximately 50 mm deep at the outer edge and slope to zero, 0.6 m in toward the centre. In making the cut, the excavated material shall be thrown to the shoulder in a small windrow against which the mixture shall be spread. The bituminous mixture shall be spread for compaction from the large windrow. It shall be bladed from this windrow in a succession of thin layers to correct thickness and a uniform cross section with a camber of 1 in 45 (except at curves, where the correct superelevation originally set for base shall be followed).

After the mixture has been spread as specified, it shall be compacted by means of rollers, which operation shall begin at the edges of the course and progress toward the centre parallel to the centreline of the roadway, overlapping in successive passes by at least one-half the width of the roller. Initial breakdown compaction may be obtained by use of steel or rubber tired rollers until no appreciable surface movement or roller marks are visible and the surface has been satisfactorily sealed. Final rolling to eliminate rubber tire or other marks and irregularities shall be carried out using an approved steel wheel finish roller or steel wheel vibrating rollers or other satisfactory equipment.

Compaction shall be carried out to the satisfaction of the Quality Manager and the Ministry Representative.

Should unsatisfactory areas develop during compaction, they shall be corrected.

521.32 Method II - When the granular base and base stabilization are included in the contract, Method II shall apply as described hereunder:

a) The first lift of granular base, 90 mm, shall be hauled to the roadway, spread, graded to line, grade and cross-section, compacted to 100% of the density as obtained in the laboratory following ASTM Designation D698 Method D.

On completion of the above roadways, as described in the foregoing, priming may be commenced.

On completion of priming and curing the second lift, 60 mm of granular base may be hauled to the primed roadway and spread in a uniform windrow and mixing commenced as described in DBSS 521.31(c).

On completion of base stabilization, the remaining granular base may be applied, i.e., from the edge of the stabilized base to the shoulder.

SECTION 536

PAVEMENT CRACK SEALING

DESCRIPTION

536.01 Scope - This work consists of restoring the asphalt concrete pavement surface to a less permeable condition. The work includes routing when necessary, cleaning and filling the crack with sealant, and dusting or sanding.

536.02 Purpose - The purpose of crack sealing shall be to prevent moisture from penetrating into the road base through surface cracks and to extend pavement life. This shall be carried out by effectively sealing the cracked surface with a crack sealant product.

MATERIALS

536.11 Asphaltic - The rubberized, asphaltic and or elasticized asphalt sealant products shall meet the requirements listed in Table 536-A.

It shall be noted that ASTM D 3405 requirements are not mandatory in the Lower Mainland, Vancouver Island or the Sunshine Coast.

As specified in the Contract the Contractor shall use a Rubberized or High Float Emulsified Asphalt HF150S or equivalent.

TABLE 536-A REQUIREMENTS

	ASTM Test	DESCRIPTION
	D 3405	Specification for joint sealants, hot poured for concrete and asphalt pavements
	D 1190	Specification for concrete joint sealer, hot poured elastic type

536.12 Sand - On pavements that require the use of High Float Emulsified Asphalt, the Contractor shall supply a "blinding sand." The sand, when tested according to ASTM C 117, shall meet the gradation requirements shown in Table 536-B and shall be approved by the Designer.

536.13 Dust Cover - Pavements that are sealed with rubberized or elasticized sealant shall receive a dust coating with a material such as Portland Cement, lime or the equivalent material approved by the Designer.

EQUIPMENT

536.21 Router - The Contractor shall provide a vertical router capable of routing asphaltic pavements to a depth of 25 mm and a width of 16 mm.

TABLE 536-B GRADATION REQUIREMENTS

Sieve (mm)	% Passing
12.5	100
9.5	80 - 100
4.75	50 - 95
2.36	30 - 80
0.600	10 - 50
0.300	0 - 25
0.075	0 - 6

536.22 Cleaner - The cleaner unit shall be a hot compressed air capable of cleaning, heating and drying routed cracks with not less than 1.7 m^3/min. of compressed air at 690 kPa. The hot air exhaust shall not exceed 315°C.

536.23 Melter - The melting kettle shall be of the double boiler type supplying indirect heating so as to slowly heat the asphaltic material with heat transfer oil. It shall have built in calibrated thermometers for both transfer oil and the sealing compound. The heating of the sealant shall be efficient and thermostatically controlled so as not to exceed the manufacturer's maximum safe heating temperature and shall be such as to maintain a constant temperature once the sealant is heated. The melter shall be capable of constantly agitating the asphaltic material as it is being heated and shall have a pump circulating the sealant from the bottom to the top of the kettle.

The melter must comply with the Gas Safety Act Regulations and Codes and any other applicable acts and regulations.

536.24 Filler Tools - Crack filler devices and strike off tools must be such that successful forming of the bead of sealant over the prepared crack is as specified in this Specification.

CONSTRUCTION

536.31 General - Crack sealing shall only be performed when the pavement surfaces are dry, and the crack and road base are dry or nearly dry (no visible moisture), and the temperature is 10°C and rising.

A random sample of virgin sealant shall be selected from each lot for testing purposes by the Quality Manager.

Cracks up to 16 mm in width shall be widened by using a router to form a sealant reservoir 16 mm in width and from 19 to 25 mm in depth. All routing shall be performed keeping the crack centreline within ± 8 mm of the centre of

the rout and shall be cleaned with a hot compressed air lance. Material removed from the cracks shall be disposed of in an environmentally sound manner.

The routing speed shall be such that the pavement is carefully cut, not broken or torn out, and the sides of the rout are smooth and uniform. The surface of the pavement and routed crack shall be cleaned of all dust and routing debris.

Routing should not be carried out on pavements that are of such an age that the pavement fractures or spalls occur along the edge of the freshly routed crack

The crack shall be filled with sealant from the bottom to the surface level in such a manner that the sealant does not bridge entrapped air pockets. Material shall be placed to overfill the crack. It will then be struck off to leave a uniform amount of sealant directly over the crack, with the edges of the spread evenly feathered to overlap on the pavement surface from a minimum of 25 mm to a maximum of 40 mm on each side of the crack. The sealant overband shall not be so thick that it can be removed during snow plowing or produce a noticeable bump when traversed by traffic.

Any damage done to the sealant by stones or any other deleterious material being embedded in the sealing compound shall be repaired.

The Contractor shall ensure that traffic is kept off the sealed cracks until such time as the sealant has properly set up and will not be damaged or pulled out by the passage of traffic.

536.32 Rubberized and Elasticized Asphalt Sealants - These shall be used on pavements that are less than 10 years old and where the majority of cracks are less than 25 mm in width.

The sealant shall be applied to cracks that have been routed to a uniform depth and width.

Cracks having a width greater than 16 mm need not be routed but shall be cleaned to a minimum depth of 25 mm.

Within two minutes of the completion of the cleaning operation the crack shall be filled with sealant from a melter using a connecting wand or manual applicator (pouring cone) which ensures minimum pour application temperatures for the product are maintained.

Upon completion of the pouring the sealant shall be dusted to prevent the asphalt from tracking. Excess dusting material shall be removed.

536.33 High Float Emulsified Asphalt - High Float Emulsified Asphalt shall be used on pavement where there are depressions or lipping at the cracks, or the majority of cracks are over 25 mm in width.

The cracks shall be cleaned as close to the actual depth as possible and the removed material shall be disposed of in an environmentally sound manner. Immediately after cleaning, they shall be filled with sealant from a distributor truck or melter. The distributor truck or melter shall have an efficient means of heating the sealant to any temperature up to 100°C and maintaining it constantly at the manufacturer's prescribed temperature without overheating.

Upon completion of the sealing of the crack, the sealant shall be sanded to prevent the asphalt from tracking. Excess sand shall be removed and disposed of in an environmentally sound manner.

SECTION 541

RUBBLE MASONRY

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

541.01 Scope - This Section covers the construction of stone paving for grouted traffic island and median capping, dry-laid and grouted slope protection, culvert and spillway aprons, ditches and other like uses.

Paving rock, thickness of 100 to 200 mm, will be specified depending upon location, use and setting requirements. Requirements for rock exceeding 200 mm minimum thickness are specified by DB SS 205 for hand-laid and grouted riprap.

MATERIALS

541.11 Materials

541.11.01 Rock - Rock for stone paving shall generally be obtained from the roadway excavation, stockpiled where designated and selected as necessary for use at the locations indicated on the Design, unless the supply of off-site rock is specified.

Rock will be required either as: angular rock as found or blasted, or rounded rock.

All rock shall be sound, durable and clean, relatively flat on the face to be exposed, reasonably consistent in colour tone, of acceptable size (200 to 450 mm across) and of the required thickness range.

541.11.02 Gravel Base - Gravel base shall be 25 mm minus well-graded crushed granular aggregate.

541.11.03 Bedding Mortar - Bedding mortar shall consist of one part Portland cement to three parts clean, well-graded fine aggregate or coarse sand mixed with water to give a 150 mm slump.

541.11.04 Setting and Jointing Mortar - Setting and jointing mortar shall consist of one part Portland cement to three parts clean, well-graded fine aggregate mixed with water to give a 75 mm slump.

CONSTRUCTION

541.31 Preparation - Graded surfaces to receive paving shall be trimmed and well tamped to the lines and grades indicated on the Design and to the approval of the Quality Manager.

Edges of paving shall be constructed of thicker stones to form a stable foundation and protection against any undercutting in waterways.

All paved areas shall receive a minimum 150 mm layer of gravel base, well consolidated.

Gravel level inside curbs shall permit the surface of the jointing mortar to be flush with the top of the curb.

541.32 Construction - Construction shall be carried out with all material, labour, tools, equipment and incidentals supplied by the Contractor as necessary to complete all stone paving work in accordance with good work practice.

Gravel base of grouted stone paving shall be well saturated with bedding mortar slurry.

All rock shall be firmly embedded into the base material so that the voids between stones are of minimal width with the rock surfaces relatively even as indicated on SS Drawing SP541-01.

Voids of grouted paving rock shall be filled with setting and jointing mortar, 75 mm thick and, after consolidation and any necessary chinking of the larger voids, the mortar surface shall be smooth finished and sloped to drain freely.

Finished stone paving shall show a clean even appearance, free of disfiguring spalls and, where grouted, cleaned of mortar staining.

SECTION 582

CONCRETE CURB AND GUTTER AND STORM DRAINAGE

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

DESCRIPTION

582.01 Scope - This Section describes the requirements for construction of roadside curb and gutter and associated underground drainage systems. Such work shall consist of constructing and/or installing appurtenances shown or indicated on the Design.

MATERIALS

582.11 Drain Pipe - All drain pipe shall be of first quality, sound, true in form and free from defects of all kinds.

582.12 Salt Glazed Clay Sewer Pipe - Salt glazed clay sewer pipe shall be in accordance with the current ASTM Specifications.

582.13 Concrete Pipe

582.13.01 Reinforced Concrete Pipe - Reinforced concrete pipe shall have joints of the tongue and groove type and shall meet the requirements of the current specification ASTM C 76 Series.

582.13.02 Plain Concrete Pipe - Plain concrete pipe shall meet the requirements of the current specification ASTM C 14 Series.

582.13.03 Requirements - Pipe shall be legibly marked with the date of manufacture, the name or trademark of the manufacturer and by marks denoting plain concrete pipe, reinforced concrete pipe or extra strength reinforced concrete pipe as the case may be. Pipe containing elliptically placed reinforcement shall be marked on the inside of the pipe with the words "Top" or "Bottom" at the correct place to indicate the proper position when laid.

582.14 Corrugated Steel Pipe - Corrugated metal pipe shall conform to the requirements of DBSS 320.

CONSTRUCTION

582.31 Portland Cement Concrete Curb and Gutter - Combined Portland cement concrete curb and gutter shall be constructed to the line and grade shown on the Design or as otherwise directed by the Designer and will include returns at the intersections if so directed.

The combined curb and gutter shall be constructed to the standard cross-section shown on SS Drawing SP582-01.01 and to such layout plans as are attached, except where the face of the curb varies and at lanes and driveways where drop curbs are ordered.

Expansion joints 13 mm in thickness cut to the cross-section of the combined curb and gutter, composed of premoulded joint filler shall be installed at approximately 15 m intervals or such closer interval as the Designer may direct.

582.32 Catch Basins - Catch basins shall be constructed, where shown and as required in the Design or as the Designer may direct, in accordance with the intent of SS Drawings SP582-02.01 through SP582-02.05.

582.33 Manholes - Manholes shall be constructed at the locations and to invert levels shown in the Design or as the Designer may direct in accordance with the intent of SS Drawings SP582-03.01 through SP582-03.08.

582.34 Storm Drains - Storm drains shall be constructed where shown on the Design, or as directed by Designer, and in accordance with the Drawings and specifications covering the various types and attached herewith. The trench and other preparatory work shall be approved by the Designer before actual placing starts.

582.35 Placing

582.35.01 Spigot and Bell Pipe - Spigot and bell pipe shall be laid beginning at the lower end with the bell end pointing up-grade. Suitable excavation or bedding must be provided to accommodate the bell so that the pipe is supported along its full length and not at the ends alone.

582.35.02 Concrete Pipe - Concrete pipe shall be laid beginning at the lower end with the groove end pointing up-grade. Pipe with elliptical reinforcement shall be laid with the minor axis of the reinforcement as marked by the manufacturer in a vertical position.

When jointing salt glazed clay and concrete pipes, the trench shall be in a dry condition and the joints shall be cleaned and wetted before sealing with mortar. The mortar shall consist of one part of Portland cement to two parts of fine sand, mixed to the proper consistency. Sealing shall be neatly and thoroughly done and the interior of the pipe cleaned of all surplus mortar. Joints shall be kept damp with burlap or earth for at least three days after sealing.

582.35.03 Corrugated Steel Pipe - Corrugated steel pipe shall be laid beginning at the lower end with the outside laps pointing up-grade and the longitudinal joint on the side. The

separate sections shall be firmly jointed together and any metal in joints which is not thoroughly protected by galvanizing shall be coated with a suitable asphaltum paint.

582.35.04 General - The Contractor shall provide and fix at such points, as may be directed, properly painted substantial sight rails set to correct levels. At least three sight rails shall be in position for each length of sewer. Pipes shall be accurately set and laid to even gradients, concentric and in straight lines between manholes.

582.36 Special Jointing Material - Where pipe trenches are necessarily restricted in width or where unavoidable wet conditions exist, which would prevent proper normal jointing of pipes with Portland cement mortar, special prefabricated joints may be authorized by the Designer to be used

582.37 Bedding and Backfilling - The bedding of all concrete pipes shall consist of 19 mm or 25 mm crushed gravel to a minimum depth of 150 mm below the pipe. The backfill shall consist of 19 mm or 25 mm crushed gravel with a minimum cover of 150 mm and in the remainder of the trench shall consist of the natural excavated material unless such material is considered unsuitable by the Designer. Such natural material shall have a maximum size of 75 mm. All materials shall be compacted in horizontal layers not more than 200 mm.

582.38 Concrete End Walls - See DBSS 303.31.04.

SECTION 586

PLANT MIXED PORTLAND CEMENT TREATED BASE COURSE

DESCRIPTION

586.01 Preliminary and General - This Specification describes the materials, plant equipment and quality of work required for the stabilization of base course material with Portland cement.

586.02 General Description of Work - The cement treated base shall consist of prepared aggregate, Portland cement and water uniformly mixed in the proportions directed, in a central mixing plant of satisfactory type, hauled out and laid on the prepared subgrade and compacted to established line and grade.

MATERIALS

586.11 Cement - Portland cement to be supplied for the work will comply with the requirements and tests of the Canadian Standards Association CAN 3-A5-M86 "Standard Portland Cements" and subsequent revisions thereof.

586.12 Water - The water shall be clean and free from substances deleterious to the hardening of the cement stabilized base course and shall be subject to the approval of the Ministry Representative.

586.13 Aggregate - The aggregate supplied shall be composed of inert durable material uniform in quality and free from organic impurities and soft or disintegrated particles. When tested according to ASTM C 136, the aggregate shall meet the gradation requirements listed in Table 586-A.

TABLE 586-A GRADATION REQUIREMENTS FOR AGGREGATE

SIEVE SIZE (mm)	PERCENT PASSING BY MASS
25	100
19	80 - 1000
9.5	50 - 100
4.75	35 - 70
2.36	25 - 50
1.18	15 - 35
0.300	5 - 20
0.075	0 - 5

The material shall have a liquid limit not greater than twenty-five and plasticity index not greater than six.

All aggregate shall be stockpiled in horizontal layers not more than 1.0 m in thickness. The stockpile area shall be cleared of all vegetation, trees, brush, rock and other debris and a uniform ground surface prepared before the aggregate is deposited on the site. The material shall be loaded from the stockpile into the plant in such a manner as to ensure that a uniform cut of the pile is obtained. Subsequent addition of prepared aggregate to the pile shall be so organized to ensure continued uniformity of the aggregate supply to the mixing plant.

EQUIPMENT

586.21 Mixing Plant - The Portland cement treated base course shall be mixed in a stationary plant employing either continuous or batch type equipment. The plant shall be equipped with interlocked feeding and metering devices which will add the cement, water and aggregates in proportions established by the mix design to a tolerance of ± 3% by weight. The plant shall be capable of mixing the ingredients uniformly. Suitable storage shall be provided for bulk cement, and if required, bulk mineral filler. The capacity of the mixing plant shall be such that a minimum of 0.4 km of full width cement treated base can be laid per daily shift, i.e., sustained mixing capacity shall normally be not less than 250 tonnes per hour. Details of the plant and its operation shall be submitted to the Ministry Representative upon request. .

586.22 Cement Treated Base Spreader - The spreading device may be self-propelled, pushed or towed; and shall be so constructed and operated as to produce without segregation or surface tearing, a layer of material which is uniform in thickness and surface contour, either longitudinal or transverse, and free from irregularities in density. Whichever type of spreader may be used, it is mandatory that it be provided with a positive mechanical means for distributing the cement treated base transversely across the front of the cutoff mechanism or screed.

Spreading equipment which rides on the freshly spread material and produces tracks or partially compacted areas behind the cutoff blade will be acceptable provided no displacement of material or filling of the tracks occurs, and provided further that the tracks are not of such depth as to be visible after compaction is completed. The use of a motor grader or any other spreading device, which requires repeated trips over the same area, will not be permitted for spreading.

586.23 Water Distributors - Water distributors shall be the pressure type equipped with a spray bar mounting nozzles similar to those used on asphalt distributors and capable of applying the water in accurate quantities and as a uniform spray. Splash plate type distributors or those equipped with spray bars that eject fine streams of water will not be permitted. The distributor must be provided with a satisfactory accurate means of measuring the quantity of water sprayed.

CONSTRUCTION

586.31 Mixing - The mix proportions shall be determined by the Designer from samples taken at the commencement of aggregate production.

The aggregate, mineral filler and cement shall be fed into the pugmill accurately, according to the proportions required, and shall be thoroughly dry mixed. Water shall then be added in the required amount and mixing continued until uniformity is achieved.

586.32 Spreading and Compaction

586.32.01 Surface - The surface of the subgrade shall be thoroughly moistened just prior to placing the cement treated base mixture thereon and shall be kept moist but not excessively wet until covered by the mixture.

586.32.02 Cement Treated Base Mixture - The cement treated base mixture shall be hauled to the roadbed in trucks equipped with protective covers and shall then be laid in a manner to produce a layer of uniform density and cross-section and in sufficient quantity to provide a compacted base conforming to the grade and cross-section set by the Design. End dumping on the roadway followed by spreading will not be permitted. Depositing and spreading the material on the roadway shall progress continuously. At any break, including the end of each day's full width construction, a straight transverse construction joint shall be formed by cutting back into the completed work to form a true vertical face and by installing a temporary wooden header, if required by the Quality Manager.

586.32.03 Compaction - Immediately following spreading, the material shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698.

586.32.04 Compaction Equipment - The use of compacting equipment which marks, ruts, cracks or otherwise loosens or damages the surface of the cement treated base will not be permitted. The compacting equipment shall have sufficient capacity to produce the specified degree of compaction within the time limits specified.

Wherever compaction planes are formed in the cement treated base during spreading, shaping or compaction, they shall be removed by scarifying or tilling the surface to such depth and in such a manner as to thoroughly break up the compaction planes. Scarifying and pulverization may be performed by any suitable equipment - nail drag, weeder, disk harrow or rotary tiller that will satisfactorily perform the work required. The scarified material will be regraded and recompacted, to conform to design Specifications.

586.32.05 Water - Water shall be applied at the rates and in the quantities directed by the Designer. The surface of the uncompacted, partially compacted and completely compacted material shall be kept moist at all times. An adequate water supply shall be maintained at the point of compacting.

586.32.06 Speed - The speed of operations shall be such that not more than forty-five minutes shall elapse between the time of starting mixing and the time of starting compaction. Shaping and final compaction shall be completed within two hours of commencement of mixing at the plant.

586.32.07 Finished Surface - The finished surface of the compacted base shall at all points be accurate to ± 15 mm of the grade and shall not, at any point, deviate by more than 10 mm from the bottom of a 3 m straight edge laid in any direction on the surface on either side of the crown of the roadway.

586.32.08 Temperature - No construction shall be undertaken when the temperature is below 4.0^{o}C or when the conditions indicate that the temperature will fall below 4.0^{o}C in the next 24 hours or if the subgrade is frozen.

586.33 Protection and Cover

586.33.01 General - Protection and cover shall be provided by one of the methods, A or B, given below and as directed by the Designer in accordance with the Contract.

a**) Method A - Crushed Gravel Course** - 25 mm crushed gravel shall be laid not more than 24 hours after the treated base is laid. The surface of the treated base and the overlying gravel shall be kept moist by the application of water as directed by the Designer.

b**) Method B - Bituminous Seal -** As soon as possible after the cement treated base course has been finished, in accordance with preceding subsections, a bituminous seal shall be applied to the surface and sanded to protect the cement treated base course from drying out during the seven day curing period. The finished base course shall be kept continuously moist until the seal has been applied.

The bituminous sealing material shall be applied at a rate between 0.7 to 1.0 /m² or as directed by the Designer in order to give complete surface coverage without excessive runoff.

Generally, this work shall be governed by the applicable provisions of DB SS 501.31 "Spray Primer and Tack Coat". At the time of sealing, the surface of the cement treated base must be dense, free from all loose and extraneous material and shall contain sufficient moisture to prevent penetration of the bituminous material. If needed, water shall be applied in sufficient quantity to fill the surface voids of the soil cement immediately before the bituminous material is applied. The seal cover shall be effectively maintained by the Contractor during the seven day curing period.

586.33.02 Protection - Any finished portion of the base course adjacent to construction, which is travelled by equipment used in constructing an adjoining section, shall be protected in such a manner as to prevent equipment from marring or damaging the completed work.

586.33.03 Temperature - At any time when the air temperature may be expected to fall to the freezing point, during the day or the night, sufficient above protection shall be given to the cement treated base to prevent its freezing for seven days after placement and until the soil cement has hardened.

586.34 Maintenance - The Contractor shall maintain the entire base course in good condition satisfactory to the Ministry Representative from commencement of construction until it has been paved with asphalt concrete. Maintenance shall include immediate repairs to any defects that may occur and shall be repeated at the Contractor's expense as often as may be necessary to keep the section continuously intact.

SECTION 604

STEEL TRAFFIC BARRIER CONSTRUCTION

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

604.01 Scope - This Section covers the construction of roadside and median steel traffic barriers in accordance with the general layout and details indicated on Drawings of the SP312 series.

Steel beam guardrail, wood posts and accessory materials are specified by DBSS 312. Steel posts are not covered in this section. Use of steel posts with steel beam guardrail will meet the provisions in NCHRP Report 350 for Test Level (TL) 3 or higher. Installation of steel posts with steel beam will be to the manufacturer's specifications, and/or to Ministry's contract requirements. Manufacturer's steel post and steel beam product must be on, or have tentative approval to be on the Ministry's recognized products list.

Design and part number references are taken from the ARTBA Technical Bulletin #268-B, as noted in DBSS 312, unless indicated otherwise on SS Drawings 1-SP312 and 2-SP312.

604.02 Provision of Steel Barriers - Barriers of the type(s) called for shall be constructed at the locations and as shown on the Design with the materials, accessories and necessary ancillary work all in accordance with the details indicated in the Contract, Specifications, Special Provisions and/or typical standard SP Drawings.

All material will be supplied by the Contractor.

Construction shall be carried out with all labour, tools, equipment and incidentals necessary to complete all barrier work in accordance with good work practice, for a substantially supported and anchored steel barrier developing a continuous beam strength together with necessary crashworthy end treatment.

604.03 Materials - Detailed material requirements are set out in DBSS 312 and indicated on SS Drawings 1-SP312, 2-SP312 and the manufacturer's drawings. Materials not listed in this section will have to be listed in the Ministry's recognized products list, or approved by the Ministry's highway engineering branch.

Concrete for post footings shall be constructed of minimum 15 MPa Portland cement concrete in accordance with Section 218, and reinforcement, 152 x 152 MW18.2 welded steel wire fabric to CSA G30.5 and 10M deformed bar or minimum 7 mm stirrups in accordance with DBSS 412, and in accordance with the Contract requirements.

Preservative treatment for protecting field cuts and borings and for making good any superficial damage to treated wood post material, as approved by the Quality Manager, shall be creosote, pentachlorophenol or preservative salts in proper solution corresponding to the original pressure treatment for application in three separate heavy coatings.

Touch-up treatment for damaged galvanized metal surfaces shall be a heavy application of zinc rich paint to CGSB Standard 1-GP-181M Specification for Ready Mixed Zinc Rich Coating.

604.04 Wood Post Installation - Posts shall be set true to the lines, spacing, depth(s) and height(s) indicated or required. The layout, type of support and fixing treatment at ends of barriers or where adjacent to abutments and the like shall be as indicated on the Design and relevant SS Drawings 1-SP312 and 2-SP312, and/or to the direction of the Designer.

Wood posts shall be driven in place or set in augered or dug holes with necessary dampened and well tamped layers of approved backfill material. Driving will be permitted only if no damage results to shoulders and adjacent slopes. In broken rock embankments the prepunching of holes will be permitted only prior to final compaction, surfacing and paving. Construction details appertaining on exposed rock, rock with minimum overburden or on concrete shall be to the indicated requirements and/or the direction of the Designer.

All work preparatory to and for the installation of posts shall be to the direction and prior approval of the Designer and Quality Manager for each type of sub-surface condition to produce a thoroughly sound guardrail support system.

Surplus excavated material remaining after guardrail support installation shall be disposed of.

Tops of all wood posts shall be set a uniform 25 mm ± 5 mm above the line of the beam's top edge and where necessary cut to line up.

Wood posts damaged by the driving operation or other damage from their handling and installation shall be replaced at the Contractor's expense.

The tops of all wood posts and offset blocks, all post bolt holes and bored holes in posts shall receive a heavy soaking coat of the specified preservative treatment on three succeeding days.

Steel posts and beam shall be installed to the manufacturer's specifications and as specified in the Contract. Steel product damaged during installation shall be replaced by the Contractor.

604.05 Steel Beam Erection - Beam sections will normally be supplied or required for joining at 3.81 m intervals.

All punching, cutting or welding shall be shop executed except for special details in unforeseen and exceptional cases and to the prior approval of the Designer.

Sections shall have full bearing one to another at laps with splicing bolts drawn tight to 100 N·m for a continuous beam effect. Where guardrail is on a curve, the beam sections shall make close contact over the whole splice area. Shop bent beams will normally be supplied or required for curvature radii under 45 m.

Beam sections shall be bolted to posts so that the edges and centre portion(s) of beams make full bearing with each offset block or where applicable with back-up plate, post or other support. Where posts are treated with waterborne preservative salts, the bolt holes shall be filled with heavy grease before bolt insertion for corrosion protection.

Bolts shall not extend more than 15 mm beyond the fixing nut when tightened, with any excess removed and the threads burred.

Washers shall be used only where indicated on the SS Drawings of the SP312 series.

604.06 End Assemblies - End assemblies shall be carried out strictly according to the manufacturer's specifications unless specifically directed otherwise by the Designer in writing as to approved alternative methods and the like.

Note the safety preference for:

a) Continuing guardrail to shield the entire length of any hazard (as indicated on SS Drawing 1-SP312), bridge abutment and the like in place of a bolted connector (RE-8).

b) Approach transitions stiffened by means of reduced post spacing and, where necessary, the use of Thrie-beam or nested twin W-beams.

c) End assemblies to be NCHRP 350 TL-3 or higher certified.

Note: the discontinuance of the flared (spade shaped) terminal section, especially at the approach ends of guardrails.

Footings for posts shall be constructed of specified concrete and reinforcement to the minimum sizes indicated on SS Drawing 2-SP312, neatly trowelled on top to weathering slope with bullnosed edge circular on plan.

SECTION 635

ELECTRICAL AND SIGNING

Note: For any SS Drawing (SP XXX - XX) referenced in this Section refer to the like-numbered Drawings in the "Standard Specifications For Highway Construction" 2006 edition.

PART A - GENERAL

635.01 Scope - This Section covers the requirements for electrical and signing works. Temporary construction signing is covered in DBSS 194 - Traffic Control.

The maintenance of the electrical infrastructure contained within the limits of the Design Build project will be undertaken by the Contractor . The Contractor will be required to coordinate with the Ministry's Manager, Electrical Services, the commissioning and turnover of electrical equipment to the Ministry's electrical maintenance contractor or Electrical Trades Supervisor, as applicable.

The Contractor shall arrange the supply, installation and removal of temporary C-63 (Traffic Pattern Changed), C-64 (Signal Operation Changed Signs) and W-329 (New) tabs with Ministry Electrical Maintenance. The Ministry's electrical maintenance contractor, or Ministry Electrical Trades Supervisor, will remove these signs at no expense to the Contractor, if removal is not required prior to the Completion Date.

635.02 Work Regulations - All electrical work shall comply with the latest edition of the Canadian Electrical Code. In addition, any applicable bulletins published by the Province of British Columbia shall apply.

The Contractor shall also conform to all applicable regulations of the Workers' Compensation Board and if required, submit a Notice of Project Form 52E49 before commencing a Ministry project. The Contractor shall ensure compliance with the following sections:

a) A Workers' Compensation Board Form 30M33 must be completed prior to working in the vicinity of overhead power lines.

b) Notice of Construction Projects, Workers' Compensation Board Occupational Health and Safety Regulation, Section 20.2.

635.03 Electrical Permits & Inspections - All electrical work shall be performed by a Registered Electrical Contractor under the provisions of the Electrical Safety Act. The Registered Electrical Contractor shall appoint at least one Registered Representative whose qualifications shall comply with the provisions of the Electrical Safety Act. The Contractor shall provide the Ministry Electrical Representative with the name and phone number of the Registered Representative prior to starting construction.

Prior to construction the Contractor shall obtain and pay for all permits required under the provisions of the Electrical Safety Act. The Contractor shall submit a copy of all permits to the Ministry Representative prior to starting construction.

Upon completion of an installation and prior to energization, the Contractor shall advise the Designer, the Quality Manager, the Ministry Representative, and the Ministry's Manager, Electrical Services that the work is complete and ready for final inspection. The Designer and Quality Manager shall undertake a review of the installation for conformance to the design and all applicable codes and regulations. The Quality Manager will report comments to the Ministry Representative.

Once the items identified have been completed and corrected the Contractor will advise the Ministry's Manager, Electrical Services who will undertake a final check and will advise if acceptable. If acceptable "notification of completion" will be issued by the Ministry's Manager, Electrical Services. If not, corrections will be required by the Contractor.

The Ministry will not accept the installation until all work has been approved by the British Columbia Safety Authority, and the Ministry's Manager, Electrical Services.

635.04 Electrical Energy Supply - The electrical energy will be supplied from overhead lines of the utility company's secondary distribution system unless otherwise noted on the Design.

The Contractor shall be responsible for making the necessary arrangements with the utility company for the connection of new service(s) and, if required, the disconnection of existing service(s). The Contractor will pay all utility connection costs.

635.05 Materials - The Contractor shall supply all materials necessary for the satisfactory completion of the project other than those materials listed in the Contract as supplied by the Ministry. All materials shall be new and conform to the requirements of the Design, the DB Specifications, and the *Electrical and Signing Material Standards Manual*.

The *Electrical and Signing Material Standards Manual* may be purchased from:

Government of British Columbia

Ministry of Labour and Citizens' Services
Publication Index Website:

http://www.publications.gov.bc.ca/

(Go Shopping)
(Keywords: *Electrical and Signing Material Standards Manual)*

email: QPPublications@gov.bc.ca
Phone: (250)387-6409 or 1-800-663-6105
Fax: (250)387-1120

Address: PO BOX 9452 STN PROV GOVT
VICTORIA BC V8W 9V7

The Electrical and Signing Material Standards is available electronically on the Ministry of Transportation web at

http://www.th.gov.bc.ca/publications/eng publications/electrical/electrical_and_traffic eng/2003_material_standards/2003_material standards.htm

All products contained in the Electrical and Signing Material Standards shall be supplied from the Ministry's *Recognized Products Book*. This list identifies the manufacturer, the approved product, the product model number and the product approval date. This list can be viewed via the Ministry web page at:

http://www.th.gov.bc.ca/publications/eng publications/geotech/rpb.htm

This list will be updated periodically as products become pre-approved.

All electrical materials shall conform to all applicable CSA Standards and shall meet the approval of the British Columbia Safety Authority Inspector.

Unless noted otherwise, all permanent signs shall meet current Ministry specifications. The Ministry *Specifications for Standard Highway Sign Materials, Fabrication and Supply* are located on our Ministry web page at:

http://www.th.gov.bc.ca/publications/eng _publications/ signs/Sign_Fabrication_Specs.pdf

Where alternative materials are permitted, approval must be received from the Ministry Representative prior to their use.

635.06 Utilities - Existing utilities are generally not shown on the electrical and signing Drawings. Where utilities are shown on the Drawings, their locations are approximate.

The Contractor shall locate and protect all existing utilities such as power lines, fibre optic cables, telephone lines, gas and oil pipelines, sewers, water works, etc. The Contractor shall at all times conduct its operations in accordance with the requirements of the utility authorities having jurisdiction. In the event of any damage to utilities, the Contractor shall be held responsible for the cost of all necessary repairs and restoration to the satisfaction of the Ministry Representative and Utility Authority.

PART B - UNDERGROUND

635.07 Concrete Bases - Concrete bases shall be constructed in accordance with Drawings SS SP635-1.1.1 through SS SP635-1.1.43 and SS SP635-1.4.1 through SS SP635-1.4.5. The use of the term "concrete bases" in DBSS 635 shall also mean "concrete spread footings" where applicable.

The Design makes reference to concrete bases that are poured in place or precast. Poured-in-place bases shall be constructed in the excavation whereas precast bases shall be constructed outside the excavation (e.g., at a precast plant or precast on site by the Contractor). The Contractor has the option to use either precast or poured- in-place bases.

Supply and installation of concrete bases shall include the following:

- excavation and backfill;
- supply and installation of concrete and formwork;
- supply and installation of reinforcing steel;
- supply and installation of conduit;
- supply and installation of steel tubing sleeves in sign post bases, and
- installation of anchor bolts.

For poured-in-place bases, the Contractor shall notify the Ministry Representative prior to pouring concrete.

The Contractor shall check for conflicts with overhead lines prior to excavating for concrete bases. If it appears there may be an overhead conflict, the Contractor shall contact the Designer for further instructions. If the Contractor installs a concrete base in a location where the pole conflicts with overhead power lines, the Contractor shall remove the pole and relocate the concrete base to a location approved by the Designer.

635.07.01 Excavation and Backfill - Excavations shall meet the requirements of DBSS 407 - Foundation Excavation.

All excavation work shall be carried out as required to suit concrete bases. Where directed by the Designer, excavations shall be shored to avoid the cutting of pavement, sidewalks, and curb and gutter.

Base excavations shall be backfilled using the excavated material provided it consists of clean well graded granular soil having a maximum fines content of 8% (silt and clay size particles) and a maximum aggregate size of 100 mm. Where backfill material does not conform as noted above, base excavation shall be backfilled with 25 mm Well Graded Base Course Aggregate material as specified in DBSS 635.14. Unacceptable materials shall be disposed of as specified in DBSS 635.15.

Backfill shall be placed in layers not exceeding 150 mm compacted thickness and shall be compacted to a minimum 100% of the laboratory density obtained by the current ASTM test method D 698. Layer thickness shall be reduced and moisture content of the material adjusted as required to achieve compaction. Care shall be taken not to damage conduits.

All excavated and backfilled areas shall be restored to their original condition.

Asphalt restoration shall be performed in accordance with DBSS 635.12.

Concrete curb and gutter restoration shall be performed in accordance with DBSS 635.13.

635.07.02 Concrete and Formwork - Concrete construction, including formwork, shall meet the requirements of DBSS 211 - Portland Cement Concrete.

Proportioning of the concrete mixes shall be the responsibility of the Contractor. The Contractor shall notify the Ministry Representative 48 hours prior to pouring concrete for poured-in-place bases.

The concrete mix shall meet the specifications listed in Table 635-A.

Concrete shall be vibrated in accordance with DBSS 211 - Portland Cement Concrete.

All exposed concrete surfaces shall be given a Class 3 finish and all buried surfaces a Class 1 finish in accordance with DBSS 211.17.

When supplying poured-in-place concrete bases, the Contractor shall have the concrete strength verified, using concrete test cylinders, prior to installing the structure onto the base. Contrary to DBSS 211 the minimum number of concrete strength tests shall be one each week per mix design or concrete supplier. The Contractor shall take concrete test cylinders at the location of the pour and perform tests in accordance with DBSS 211.09.01. The Contractor must provide written confirmation of concrete test cylinder results prior to installing any structure on the bases.

TABLE 635-A CONCRETE MIX

Minimum compressive strength at 28 days	30 MPa
Maximum nominal aggregate size	28 mm
Maximum W/C ratio by mass	0.45
Air content	5 ± 1%
Slump	50 ± 20 mm

Where installations are of a fast-track nature, the Contractor shall use a stronger concrete mix that will give the equivalent 28 day strength.

All concrete bases shall have their concrete strength verified with the exception of controller bases, sign post bases and post mounted flasher bases. Pre-cast concrete bases constructed off-site shall have their strength verified by the supplier.

The top finished surface of a concrete base shall not vary by more than 3 mm in depth as measured across the widest surface. If this variation exceeds 3 mm then the top 50 mm of the base shall be broken off by hand and reformed or the base shall be replaced.

635.07.03 Reinforcing Steel - Reinforcing steel shall meet the requirements of DBSS 412 - Reinforcing Steel.

All reinforcing steel shall conform to CAN/CSA-G30.18-M 400R.

The Contractor shall adjust the spacing of reinforcing steel to suit anchor bolts and conduit.

635.07.04 Anchor Bolts and Conduit - Anchor bolts and conduits shall be cast into the concrete bases.

Where an anchor bolt is damaged it shall be repaired in accordance with Drawings SS SP635-1.1.42 and 1.1.43.

Welding of anchor bolts shall not be permitted.

635.07.05 Steel Pipe Sleeves - Steel pipe sleeves shall be cast into the concrete bases. The type of sleeve shall be as noted on the Design and shall be in accordance with DBSS 635.28 or 635.29

635.07.06 Not Used

635.08 Junction Boxes and Vaults - Plastic junction boxes shall be installed in accordance with Drawings SS SP635-1.2.1 through SS SP635-1.2.16 and SS SP635-1.4.1 through SS SP635-1.4.3.

Concrete junction boxes shall be supplied and installed in accordance with Drawings SS SP635-1.3.1, 1.3.2, and 1.3.4. Concrete vaults shall be supplied and installed in accordance with Drawings SS SP635-1.3.3 and 1.3.4.

Concrete Junction Boxes shall be designed to withstand 5000 kg static loading.

Concrete vaults shall be designed to withstand CS600 (H-20) Static Loading.

The Ministry-approved concrete junction box and vault suppliers are listed on Drawings SS SP635-1.3.1 and 1.3.3. Concrete junction box or vault suppliers wishing to seek Ministry approval must submit shop drawings sealed by a Structural Engineer (registered with the APEGBC) to the Ministry of Transportation - Engineering Branch, Victoria, verifying that their junction box or vault meets the loading requirements. The junction box or vault shall also meet the size and functional characteristics shown on the Drawings and the Design.

Excavations for junction boxes and vaults shall meet the requirements of DBSS 407 - Foundation Excavation.

All excavation work shall be carried out as required to suit junction boxes and vaults. Where possible, excavations shall be shored to avoid the cutting of pavement, sidewalks and curb and gutter.

Junction boxes and vaults shall be installed on a bedding of 25 mm Well Graded Base Course Aggregate material in accordance with DBSS 202 - Granular Surfacing, Base and Sub-bases.

Junction box or vault excavations shall be backfilled using the excavated material except where excavated material is ruled unacceptable by the Designer or Quality Manager. Where new material is required, 25 mm Well Graded Base Course Aggregate material shall be used as specified in DBSS 635.14. Unacceptable materials shall be disposed of as specified in DBSS 635.15.

Bedding and backfill material shall be placed in layers not exceeding 150 mm compacted thickness and shall be compacted to a minimum 95% of the laboratory density obtained by the current ASTM test method D 698. Layer thickness shall be reduced and moisture content of the material adjusted as required to achieve compaction. Care shall be taken not to damage conduits.

The top of junction boxes and vaults shall be installed so that they are flush with the finished grade.

Concrete for junction box collars shall conform to DBSS 635.07.02.

All areas where excavation and backfilling have been performed shall be restored to their original condition.

Asphalt restoration shall be performed in accordance with DBSS 635.12.

Concrete curb and gutter restoration shall be performed in accordance with DBSS 635.13.

Lid hold down bolts shall be coated with anti-seize lubricant.

635.09 Conduits - All electrical conduits shall be installed in accordance with the Design.

Rigid PVC (RPVC) conduit shall be unplasticized polyvinyl chloride and conform to CSA C22.2 No. 211.2. Couplings, adapters, bends and fittings shall be unplasticized polyvinyl chloride and conform to CSA C22.2 No. 85. RPVC conduit shall be installed using CSA certified cement. Each standard length of RPVC conduit and fitting shall bear a certification mark to the applicable CSA standard.

Rigid metal conduit (RMC) shall be hot-dipped galvanized rigid steel and conform to CSA C22.2 No. 45. Any exposed metal on conduit shall be coated with cold galvanizing compound in accordance with DBSS 635.23.

Flexible liquid-tight non-metallic conduit (FC) shall conform to CSA 22.2 No. 227.2. All joints shall be made with FC threaded couplers, adapters or conduit fittings.

Liquid-tight flexible metal conduit (FMC) shall conform to CSA 22.2 No. 56. All joints shall be made with FMC threaded couplers, adapters or conduit fittings.

Conduit straps shall be galvanized steel single hole or double hole type (sized to suit conduit).

All empty conduits shall have a pull string and shall be capped in accordance with Drawings SS SP635-1.2.16 and 1.3.4. Pull string shall be polypropylene with a minimum tensile strength of 1.1 kN.

The Contractor shall lay out conduit so that 90° bends are avoided. Where 90° bend cannot be avoided, only one will be allowed for each conduit run.

Conduit couplings or bell ends damaged when pulling

conductors shall be rejected and replaced at the Contractor's expense. An appropriate pulling jig or method shall be used to avoid damaging conduit or fittings when pulling conductors.

635.10 Trenching and Backfilling for Conduit - Trenching and backfilling shall be performed in accordance with Drawings SS SP635-1.5.1, 1.5.2, 1.6.1, 1.6.2 and 1.7.1.

RPVC conduits shall be installed in an open trench in accordance with Drawings Ss SP635-1.5.1, 1.5.2, 1.6.1, 1.6.2 and 1.7.1 except where conduits are to be drilled or pushed under the surface as specified in DBSS 635.11.

Trenches shall be excavated with neat, uniform sides to the minimum width necessary, but not less than the minimum dimensions indicated on Drawings SS SP635-1.5.1 and 1.5.2 (care shall be taken to prevent sloughing). Material in the floor of the trench shall be undisturbed, or if disturbed shall be re-compacted to the requirement herein for backfill.

Trenches shall be backfilled using the excavated material except where excavated material is ruled unacceptable for backfill. Where new material is required, 25 mm Well Graded Base Course Aggregate material shall be used as specified in DBSS 635.14. Unacceptable materials shall be disposed of as specified in DBSS 635.15.

A 150 mm wide yellow plastic marker tape indicating "WARNING ELECTRICAL" shall be installed in each trench for the complete length of the trench. The location of the marker tape shall be as shown on Drawings SS SP635-1.5.1 and 1.5.2.

Bedding and backfill material shall be placed in layers not exceeding 150 mm compacted thickness and shall be compacted to a minimum 100% (for road crossings) and 95% (for shoulder trenches) of the laboratory density obtained by the current ASTM test method D 698. Layer thickness shall be reduced and moisture content of the material adjusted as required to achieve compaction. Care shall be taken not to damage conduits.

All areas where trenching and backfilling have been performed shall be restored to their original condition.

Asphalt restoration shall be performed in accordance with DBSS 635.12.

Concrete curb and gutter and sidewalk restoration shall be performed in accordance with DBSS 635.13.

635.11 Trenchless Conduit Installation - Where noted on the Design the Contractor shall install the conduit using a suitable trenchless technology to avoid cutting the existing pavement or sidewalk.

The trenchless technology selected shall suit the site conditions.

The Contractor shall give the Ministry Representative a minimum of 48 hours notice prior to installing conduits.

The Contractor shall locate all underground utilities prior to installing conduits.

635.12 Asphalt Pavement - Asphalt paving shall be performed in accordance with DBSS 501 - Hot Mixed Asphalt Pavement.

Asphalt patching of conduit trenches shall be performed in accordance with Drawing SS SP635-1.5.1. Asphalt patching of excavations for junction boxes, vaults and bases shall be performed similar to the details shown on Drawing SS SP635-1.5.1.

Contrary to the requirements of DBSS 501, Asphalt placing by a paving machine will not be required. Asphalt pavement restoration shall be Class 1 medium mix installed to the depth shown in the Design or to match pre-existing conditions, installed in lifts of a maximum of 50 mm.

The Contractor shall use a pavement saw to cut pavement and shall exercise care to obtain a neat, uniform excavation line along the edge of the pavement. The Contractor shall also prevent the existing base gravel from sloughing and undermining the pavement.

Where three lifts of asphalt pavement are being placed to reinstate a trench, the first two lifts shall be compacted with vibratory equipment capable of operating within the excavated area and of compacting over the whole of it. The third lift shall be compacted with rolling equipment to produce a uniform surface matching the adjacent existing pavement.

635.13 Concrete Curbs and Sidewalks - All concrete curbs and sidewalks shall be installed in accordance with DBSS 582 - Concrete Curb and Gutter and Storm Drainage.

Concrete shall meet the requirements of Note 1 on Drawing SS SP582-01.01.

The Contractor shall use a pavement saw to cut existing concrete and shall obtain a neat, uniform cut along the edge of concrete. The Contractor shall also prevent base gravel from sloughing and undermining the concrete.

635.14 25 mm Well Graded Base Course Aggregate - 25 mm Well Graded Base Course Aggregate shall meet the

requirements of DBSS 202 - Granular Surfacing, Base and Sub-bases.

When acceptable excavated material is not available, 25 mm Well Graded Base Course Aggregate shall be used as backfill. It shall also be used for additional backfill around bases, junction boxes, vaults, median island fill and other work.

635.15 Removal of Rejected Excavated Material - Where excavated material is designated unacceptable for backfill it shall be removed from the work site and disposed of by the Contractor at an approved site.

635.16 Restoration - Areas where work has been performed shall be returned to their original condition and must be left in a neat state to the satisfaction of the Quality Manager and Ministry Representative.

PART C - ELECTRICAL

635.17 Luminaire, Signal and Sign Poles - Luminaire and signal poles shall be installed in accordance with Drawings SSSP635-2.1.1 through SS SP635-2.1.14, and SS SP635-2.2.1 through SS SP635-2.2.10.

Where specified, luminaire poles shall be mounted on frangible or breakaway bases in accordance with Drawings SS SP635-2.1.15 and 2.1.16.

Sign poles shall be installed in accordance with Drawings SS SP635-3.1.1 through SS SP635-3.1.22.

Poles shall not be erected until concrete bases have attained a strength of 30 MPa.

Push button and signal head locations shall be verified before poles are drilled and assembled.

Field drilling of holes larger than 33 mm (1 5/16") diameter will not be permitted in Type 1, 3, 6, 7, L, M and H shafts, and all arms and extensions. Where larger holes are required, the holes shall be reinforced with a welded bushing in accordance with Drawing SS SP635-2.4.13.

All components of the luminaire, signal and sign poles shall be handled with care to prevent stress to the components through bending or twisting. A nylon sling shall be used to transport and erect the components. The use of steel chains as slings will not be permitted. Any damage to the components through overstress, scratching or denting shall be repaired or replaced at the Contractor's expense to the satisfaction of the Quality Manager and Ministry Representative.

All pole shafts shall be installed plumb. Where possible, all luminaire poles shall be positioned with the handhole oriented opposite the road and if not possible, on the downstream traffic side.

All wiring inside the poles shall be in accordance with DBSS 635.19.

Pole shafts shall mount directly on concrete bases and, where necessary, shall be trued to plumb using levelling shims.

The Contractor shall tighten all bolts and nuts to 1/3 past snug tight. "Snug-tight" is the tightness attained by a few impacts of an impact wrench or the full effort of a person using a spud wrench.

All scratches in poles and all field drilled holes shall be coated with cold galvanizing compound as per DBSS 635.23.

Hand hole bolts shall be coated with anti-seize lubricant.

635.17.01 Traffic and Pedestrian Signal Heads - Traffic and pedestrian signal heads shall be installed in accordance with Drawings SS SP635-2.3.1 through SS SP635-2.3.6 and -2.3.10.

All traffic and pedestrian signal heads and mounting hardware shall be securely attached to the pole.

Traffic and pedestrian signal heads shall be correctly aimed in the field to the satisfaction of the Designer and the Ministry Representative.

All traffic and pedestrian signal heads shall have LED (light emitting diode) traffic signal heads unless noted otherwise on the Contract.

The Contractor shall completely cover all traffic and pedestrian signal heads with dark coloured pre manufactured signal cover bags from the time they are installed until the system is activated.

The Contractor shall aim all traffic signal heads in accordance with Drawing SS SP635-2.3.11. Pedestrian signal heads shall be aimed to line up with the opposing wheelchair ramp or centre of the crosswalk.

635.17.02 Pedestrian Pushbuttons and Signs - Pedestrian pushbuttons and signs shall be installed in accordance with Drawing SS SP635-2.3.7.